



Fair, Accountable, Transparent Machine Learning (FAccT ML)

ZG517



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Session 4

Time – 10:40 AM to 12:40 PM

These slides are prepared by the instructor, with grateful acknowledgement of many others who made their course materials freely available online.

Fair ML Methods



- Pre-processing Methods
 - Transform data before ML models learn
 - e.g., Reweighting, Resampling
- In-processing Methods
 - Constrain ML models while they learn, e.g.,
 - Prejudice Removing Regularizer
 - Adversarial Learning
- Post-processing Methods
 - Make predictions from a black-box ML model fair in the post-processing stage
 - e.g., Learning to Defer

Session Content



- In-processing Methods
 - Prejudice Preserving Regularization
 - Adversarial Learning
- Post-processing Methods
 - Learning to Defer

Prejudice Remover Regularizer

Quantified causes of unfairness



Prejudice

- Unfairness rooted in the dataset

Underestimation

- Model unfairness because the model is not fully converged

Negative Legacy

- Unfairness due to sampling biases

Training Objective

$$- \underbrace{\mathcal{L}(\mathcal{D}; \Theta)}_{\text{Loss of the Model}} + \underbrace{\eta R(\mathcal{D}, \Theta)}_{\text{Fairness Regularizer}} + \underbrace{\frac{\lambda}{2} \|\Theta\|_2^2}_{\text{L2 Regularizer}}$$

[Kamishima et al, 2012](#)

Prejudice Index (PI)

Recall that Indirect Discrimination Happens When



Prediction is not directly conditioned on sensitive variables S

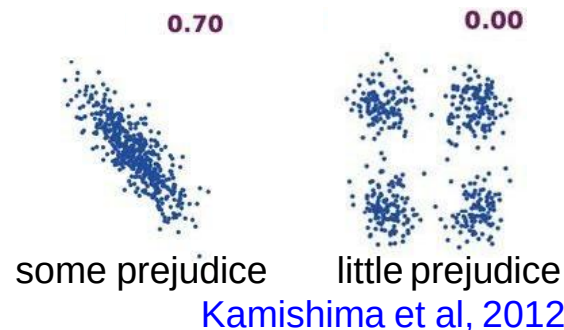
Prediction is indirectly conditioned on S by a variable O that is dependent on S

Prejudice Index (PI)

- Measures the degree of indirect discrimination based on mutual information

$$PI = \sum_{(y,s) \in \mathcal{D}} \hat{P}_r[y, s] \ln \frac{\hat{P}_r[y, s]}{\hat{P}_r[y] \hat{P}_r[s]}$$

↑
prediction model



[Kamishima et al, 2012](#)

Normalized Prejudice Index (NPI)

- Prejudice Index (PI)

- Measures the degree of indirect discrimination based on mutual information
- Ranges in $[0, +\infty)$

$$PI = \sum_{(y,s) \in \mathcal{D}} \hat{Pr}[y, s] \ln \frac{\hat{Pr}[y, s]}{\hat{Pr}[y] \hat{Pr}[s]}$$

- Normalized Prejudice Index (NPI)

- Normalize PI by the entropy of Y and S
- Ranges in $[0, 1]$

$$NPI = PI / (\sqrt{H(Y)H(S)})$$

Optimizing PI



- Learning PI

$$PI = \sum_{Y,S} \hat{Pr}[Y, S] \ln \frac{\hat{Pr}[Y, S]}{\hat{Pr}[S] \hat{Pr}[Y]}$$

Optimizing PI



- Learning PI

Expands $\Pr(Y, S)$ into $\sum_x \Pr(X, Y, S)$

$$\text{PI} = \sum_{Y, S} \hat{\Pr}[Y, S] \ln \frac{\hat{\Pr}[Y, S]}{\hat{\Pr}[S] \hat{\Pr}[Y]} = \sum_{X, S} \tilde{\Pr}[X, S] \sum_Y \mathcal{M}[Y|X, S; \Theta] \ln \frac{\hat{\Pr}[Y, S]}{\hat{\Pr}[S] \hat{\Pr}[Y]}.$$

double summations triple summations Prediction Model

- Using Logistic Regression Model as the Prediction Model

$$\mathcal{M}[y|\mathbf{x}, s; \Theta] = y\sigma(\mathbf{x}^\top \mathbf{w}_s) + (1 - y)(1 - \sigma(\mathbf{x}^\top \mathbf{w}_s))$$

Optimizing PI



- Learning PI

$$\begin{aligned} \text{PI} &= \sum_{Y,S} \hat{\text{Pr}}[Y, S] \ln \frac{\hat{\text{Pr}}[Y, S]}{\hat{\text{Pr}}[S] \hat{\text{Pr}}[Y]} = \sum_{X,S} \tilde{\text{Pr}}[X, S] \sum_Y \mathcal{M}[Y|X, S; \boldsymbol{\Theta}] \ln \frac{\hat{\text{Pr}}[Y, S]}{\hat{\text{Pr}}[S] \hat{\text{Pr}}[Y]} \\ &= \sum_{(\mathbf{x}_i, s_i) \in \mathcal{D}} \sum_{y \in \{0,1\}} \mathcal{M}[y|\mathbf{x}_i, s_i; \boldsymbol{\Theta}] \ln \frac{\hat{\text{Pr}}[y|s_i]}{\hat{\text{Pr}}[y]}. \end{aligned}$$

- Using Logistic Regression Model as the Prediction Model

difficult to evaluate

$$\mathcal{M}[y|\mathbf{x}, s; \boldsymbol{\Theta}] = y\sigma(\mathbf{x}^\top \mathbf{w}_s) + (1 - y)(1 - \sigma(\mathbf{x}^\top \mathbf{w}_s))$$

Optimizing PI



$$\text{PI} = \sum_{(\mathbf{x}_i, s_i) \in \mathcal{D}} \sum_{y \in \{0,1\}} \mathcal{M}[y | \mathbf{x}_i, s_i; \Theta] \ln \frac{\hat{\text{Pr}}[y | s_i]}{\hat{\text{Pr}}[y]}$$

[Kamishima et al, 2012](#)

Optimizing PI



$$\text{PI} = \sum_{(\mathbf{x}_i, s_i) \in \mathcal{D}} \sum_{y \in \{0,1\}} \mathcal{M}[y|\mathbf{x}_i, s_i; \boldsymbol{\Theta}] \ln \frac{\hat{\text{Pr}}[y|s_i]}{\hat{\text{Pr}}[y]}$$

$$\hat{\text{Pr}}[y|s] = \int_{\text{dom}(X)} \text{Pr}^*[X|s] \mathcal{M}[y|X, s; \boldsymbol{\Theta}] dX$$

Integrals Are Difficult to Evaluate

Optimizing PI



$$\text{PI} = \sum_{(\mathbf{x}_i, s_i) \in \mathcal{D}} \sum_{y \in \{0,1\}} \mathcal{M}[y|\mathbf{x}_i, s_i; \boldsymbol{\Theta}] \ln \frac{\hat{\text{Pr}}[y|s_i]}{\hat{\text{Pr}}[y]}$$

$$\begin{aligned} \hat{\text{Pr}}[y|s] &= \int_{\text{dom}(X)} \text{Pr}^*[X|s] \mathcal{M}[y|X, s; \boldsymbol{\Theta}] dX \\ &\approx \frac{\sum_{(\mathbf{x}_i, s_i) \in \mathcal{D} \text{ s.t. } s_i = s} \mathcal{M}[y|\mathbf{x}_i, s; \boldsymbol{\Theta}]}{|\{(\mathbf{x}_i, s_i) \in \mathcal{D} \text{ s.t. } s_i = s\}|} \end{aligned}$$

Approximating integrals by sample means

Optimizing PI



$$\text{PI} = \sum_{(\mathbf{x}_i, s_i) \in \mathcal{D}} \sum_{y \in \{0,1\}} \mathcal{M}[y|\mathbf{x}_i, s_i; \boldsymbol{\Theta}] \ln \frac{\hat{\text{Pr}}[y|s_i]}{\hat{\text{Pr}}[y]}$$

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Approximating integrals by sample means

Optimizing PI



$$\text{PI} = \sum_{(\mathbf{x}_i, s_i) \in \mathcal{D}} \sum_{y \in \{0,1\}} \mathcal{M}[y|\mathbf{x}_i, s_i; \Theta] \ln \frac{\hat{\text{Pr}}[y|s_i]}{\hat{\text{Pr}}[y]}$$

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Approximating integrals by sample means

$$\hat{\text{Pr}}[y] \approx \frac{\sum_{(\mathbf{x}_i, s_i) \in \mathcal{D}} \mathcal{M}[y|\mathbf{x}_i, s_i; \Theta]}{|\mathcal{D}|}$$

Optimizing PI



$$\text{PI} = \sum_{(\mathbf{x}_i, s_i) \in \mathcal{D}} \sum_{y \in \{0,1\}} \mathcal{M}[y|\mathbf{x}_i, s_i; \boldsymbol{\Theta}] \ln \frac{\hat{\text{Pr}}[y|s_i]}{\hat{\text{Pr}}[y]}$$

$$\hat{\text{Pr}}[y|s] \approx \frac{\sum_{(\mathbf{x}_i, s_i) \in \mathcal{D} \text{ s.t. } s_i=s} \mathcal{M}[y|\mathbf{x}_i, s; \boldsymbol{\Theta}]}{|\{(\mathbf{x}_i, s_i) \in \mathcal{D} \text{ s.t. } s_i = s\}|} \quad \hat{\text{Pr}}[y] \approx \frac{\sum_{(\mathbf{x}_i, s_i) \in \mathcal{D}} \mathcal{M}[y|\mathbf{x}_i, s_i; \boldsymbol{\Theta}]}{|\mathcal{D}|}$$

[Kamishima et al, 2012](#)

Putting Things Together



Optimization Target

$$- \mathcal{L}(\mathcal{D}; \Theta) + \eta R(\mathcal{D}, \Theta) + \frac{\lambda}{2} \|\Theta\|_2^2$$

Loss of the Model

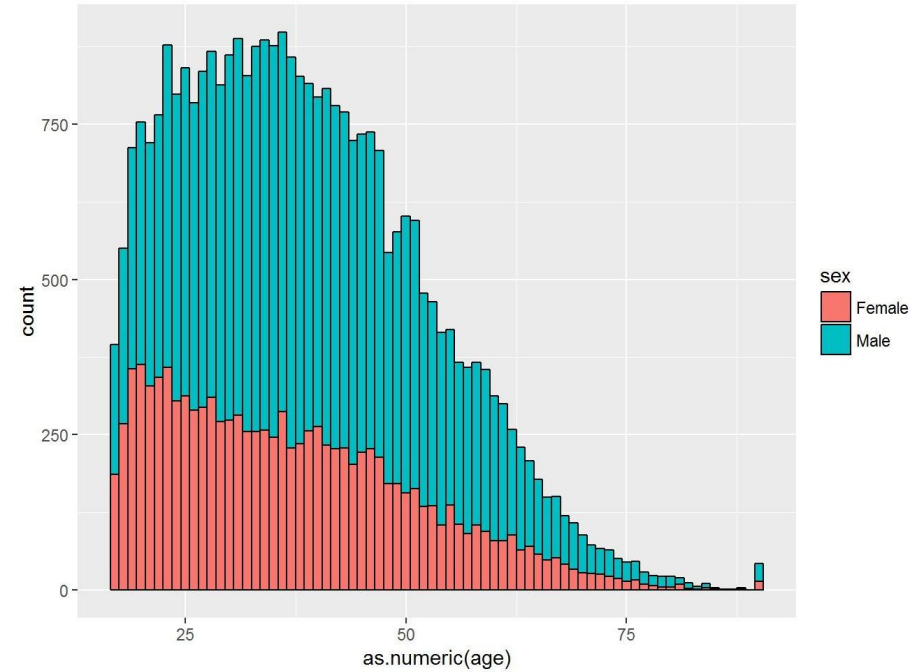
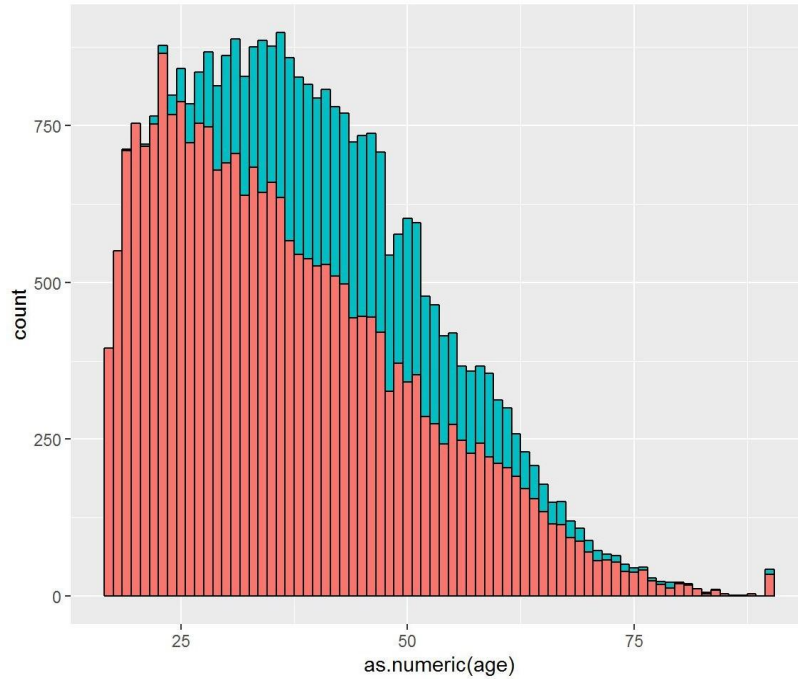
Fairness Regularizer

L2 Regularizer

- Fairness Regularizer

$$\text{PI} = \sum_{(\mathbf{x}_i, s_i) \in \mathcal{D}} \sum_{y \in \{0,1\}} \mathcal{M}[y|\mathbf{x}_i, s_i; \Theta] \ln \frac{\hat{\text{Pr}}[y|s_i]}{\hat{\text{Pr}}[y]}$$

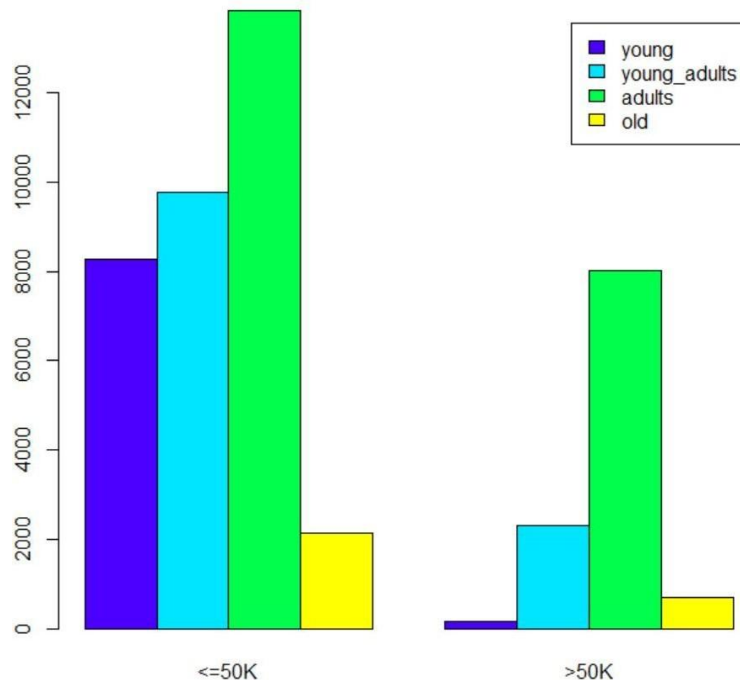
Adult Income Dataset



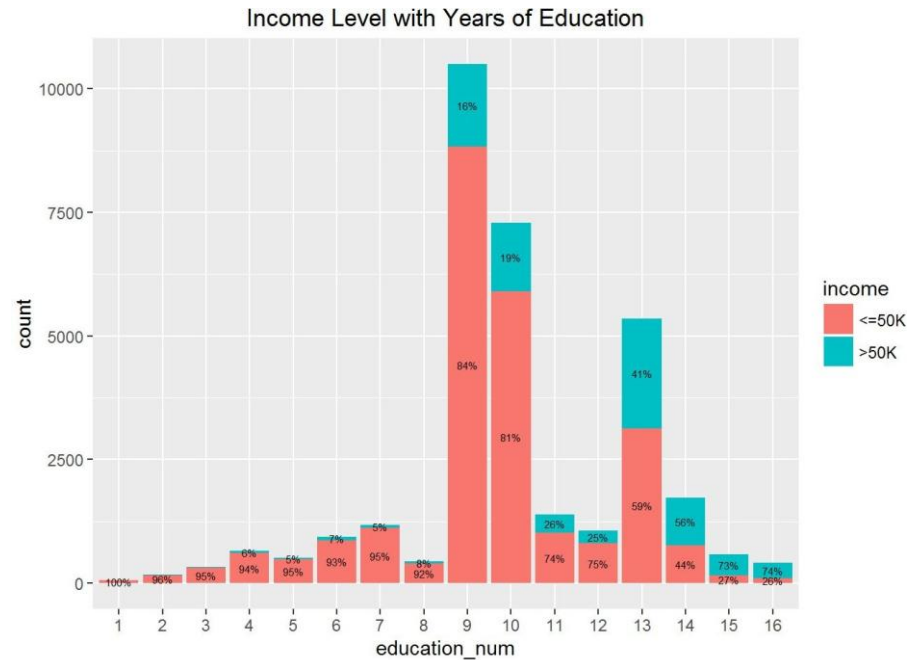
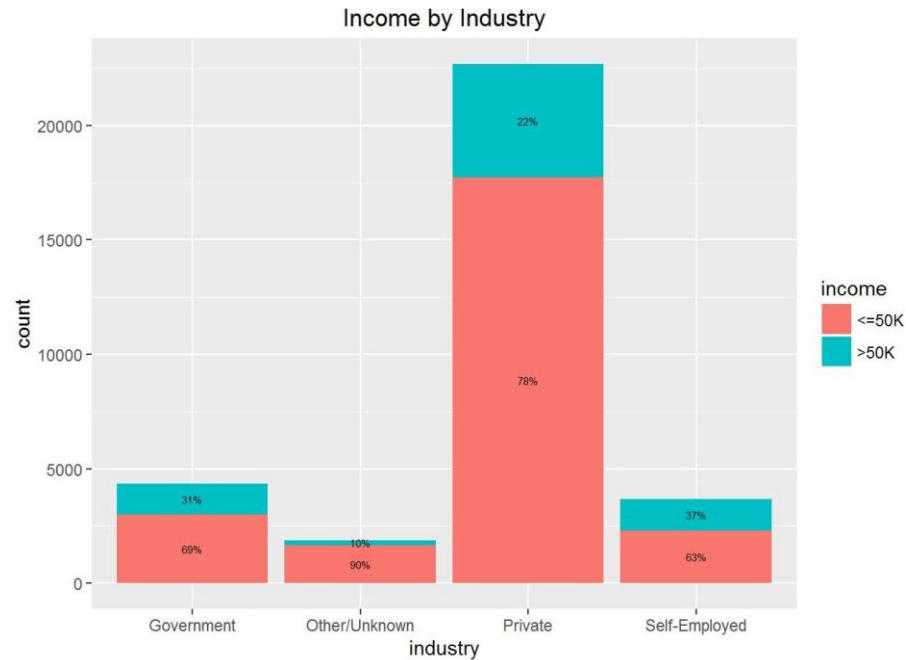
Adult Income Dataset



- Predict Whether Income Exceeds \$50K/yr Based on Census Data



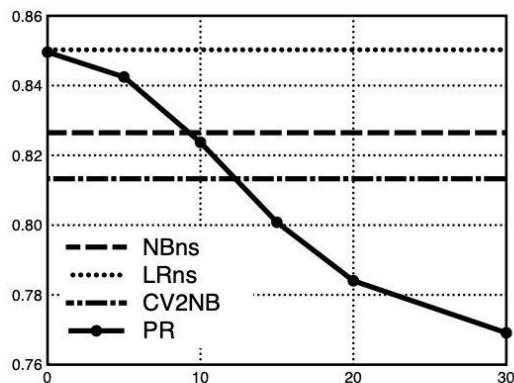
Adult Income Dataset



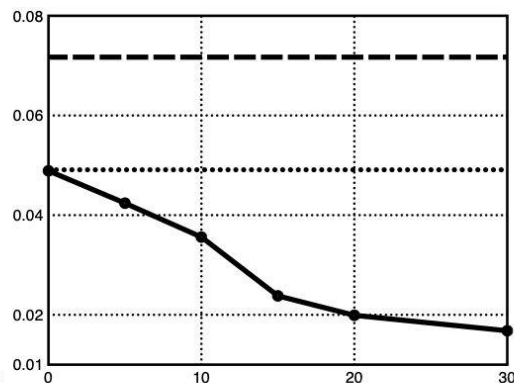
Results



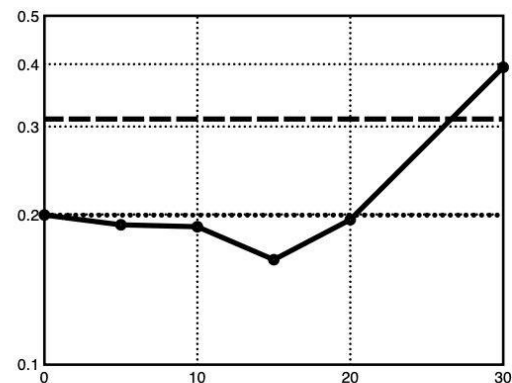
- Changes of Performance With η
 - Model performance decreases (Acc)
 - Discrimination Decreases (NPI)
 - "Fairness Efficiency" (PI/MI) Increases



(a) Acc



(b) NPI



(c) PI/MI

[Kamishima et al, 2012](#)

Results



- Prejudice Prior Sacrifices Model Performance
 - PR has lower Acc (Accuracy)
 - PR has lower NMI (normalized mutual information between labels and predictions)
- Prejudice Prior Makes Model Fair
 - PR has lower NPI

	method	Acc	NMI	NPI	PI/MI
Logistic Regression full fet.	→ LR	0.851	0.267	5.21E-02	2.10E-01
Logistic Regression no sensitive fet.	→ LRns	0.850	0.266	4.91E-02	1.99E-01
	→ PR $\eta=5$	0.842	0.240	4.24E-02	1.91E-01
Logistic Regression + Prejudice Regularizer	→ PR $\eta=15$	0.801	0.158	2.38E-02	1.62E-01
	→ PR $\eta=30$	0.769	0.046	1.68E-02	3.94E-01

η is the weight we put on prejudice regularizers [Kamishima et al, 2012](#)

Results



- PI/MI
 - Prejudice Index / Mutual Information
 - Demonstrates a trade-offs between model fairness and performance
 - Measures the amount of discrimination we eliminate with one unit of performance gain (measured by MI)

		method	Acc	NMI	NPI	PI/MI
Logistic Regression full fet.	→	LR	0.851	0.267	5.21E-02	2.10E-01
Logistic Regression no sensitive fet.	→	LRns	0.850	0.266	4.91E-02	1.99E-01
Logistic Regression + Prejudice Regularizer	↗	PR $\eta=5$	0.842	0.240	4.24E-02	1.91E-01
	→	PR $\eta=15$	0.801	0.158	2.38E-02	1.62E-01
	↘	PR $\eta=30$	0.769	0.046	1.68E-02	3.94E-01

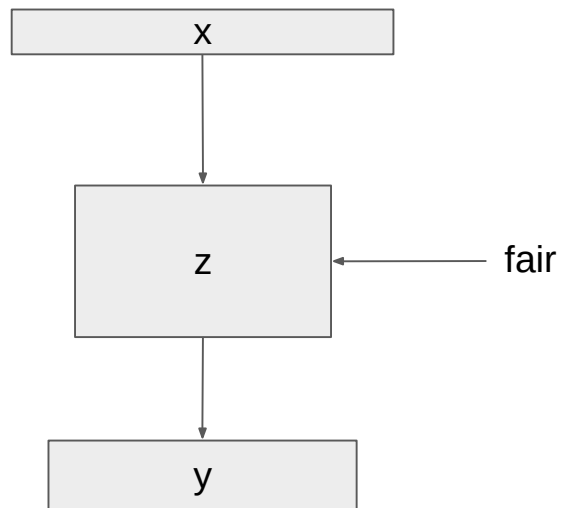
η - weight put on the prejudice regularizer

[Kamishima et al, 2012](#)

Fair Representation Learning



- Ensure fairness up to a certain level



Fair Representation Learning

- How Do We Test the Fairness of Representation Z ?
 - Adversarial Learning

I want to find the best representation for my task.



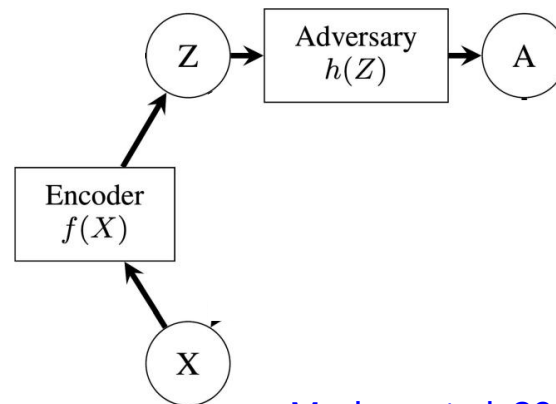
Adversarial Learning



I want to find the worst representation that can reconstruct A

Fair Representations

- How Do We Make a Representation Z Fair?
 - $Z = f(X)$
 - Test and see if a good amount of A can be reconstructed from Z
 - Compare A with $h(z)$



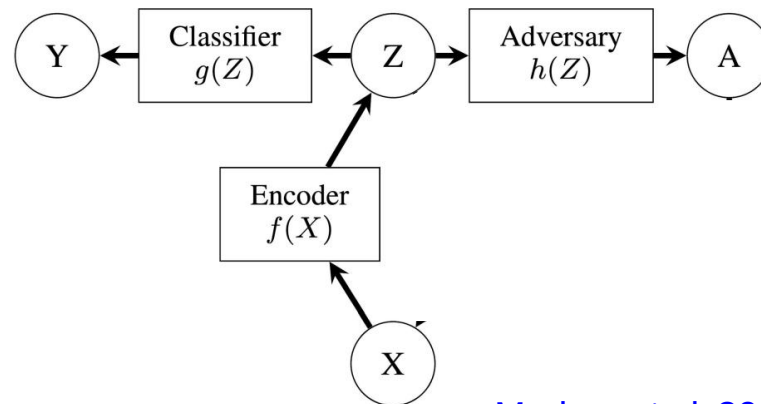
[Madras et al, 2018](#)

Fair Representations



- How Do We Make a Representation Z Fair?
 - $Z = f(X)$
 - Test and see if a good amount of A can be reconstructed from Z
 - Compare A with $h(z)$

- Properties of Deep Representations
 - Achieve good performance for downstream task that generates $y=g(z)$

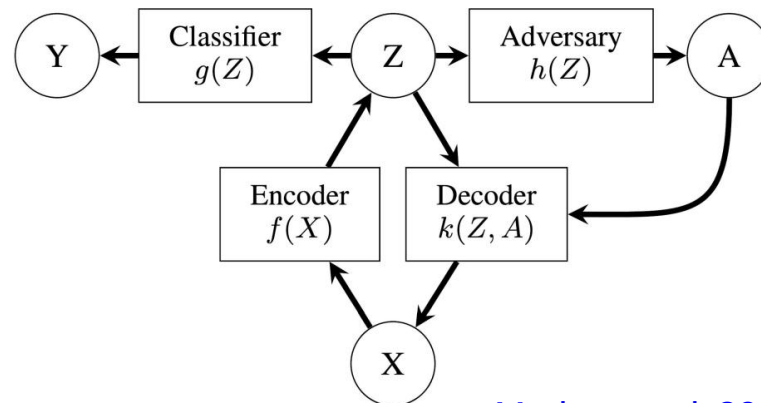


[Madras et al, 2018](#)

Fair Representations

- How Do We Make a Representation Z Fair?
 - $Z = f(X)$
 - Test and see if a good amount of A can be reconstructed from Z
 - Compare A with $h(z)$

- Properties of Deep Representations
 - Achieve good performance for downstream task that generates $y=g(z)$
 - Has the Ability to Reconstruct $X = k(Z, A)$



[Madras et al, 2018](#)

Fairness Through Adversarial Learning

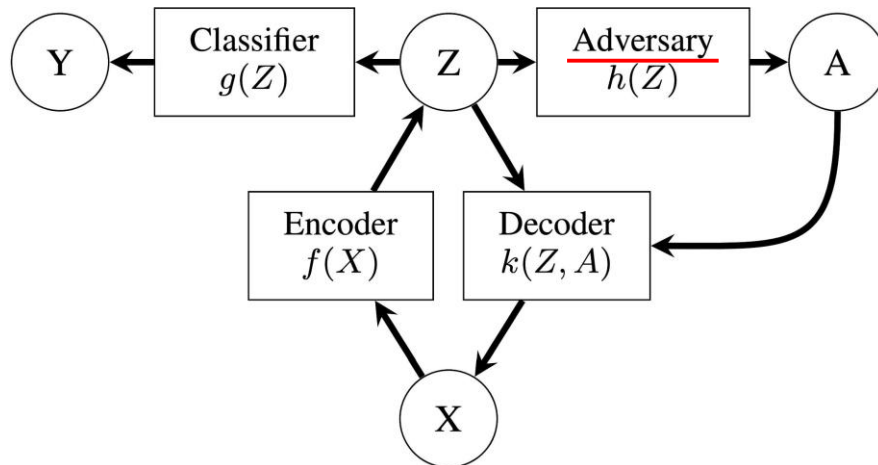
innovate

achieve

lead

- Adversarial Learning

- Models are trained using objectives that compete with each other



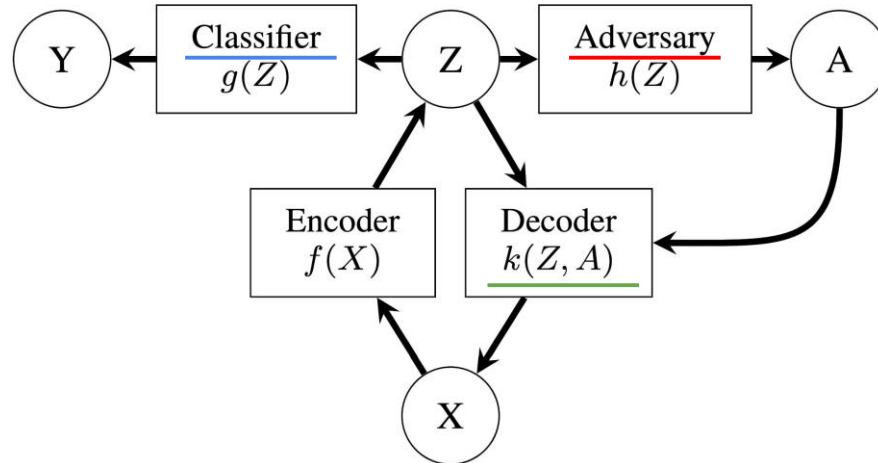
$$\underset{f,g,k}{\text{minimize}} \underset{h}{\text{maximize}} \mathbb{E}_{X,Y,A} [L(f, g, h, k)]$$

[Madras et al, 2018](#)

Fairness Through Adversarial Learning

- Adversarial Learning

$$L(f, g, h, k) = \alpha \underline{L_C(g(f(X, A)), Y)} + \beta \underline{L_{Dec}(k(f(X, A), A), X)} + \gamma \underline{L_{Adv}(h(f(X, A)), A)}$$



$$\underset{f, g, k}{\text{minimize}} \underset{h}{\text{maximize}} \mathbb{E}_{X, Y, A} [L(f, g, h, k)]$$

[Madras et al, 2018](#)

Loss for Learning Fair Representations

- Adversarial Loss for Demographic Parity with Group $\mathcal{D}_0, \mathcal{D}_1$

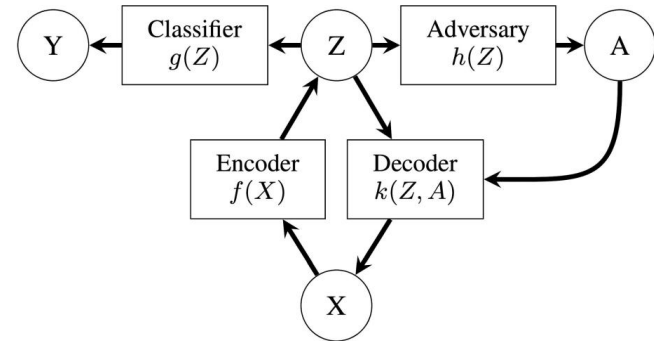
$$L_{Adv}^{DP}(h) = 1 - \sum_{i \in \{0,1\}} \frac{1}{|\mathcal{D}_i|} \sum_{(x,a) \in \mathcal{D}_i} |h(f(x,a)) - a|$$

Demographic Parity: $P(\hat{Y} = 1|A = 1) = P(\hat{Y} = 1|A = 0)$

- Adversarial Loss for Equality of Odds with Group $\mathcal{D}_i^j = \{(x, y, a) \in \mathcal{D} | a = i, y = j\}$

$$L_{Adv}^{EO}(h) = 2 - \sum_{(i,j) \in \{0,1\}^2} \frac{1}{|\mathcal{D}_i^j|} \sum_{(x,a) \in \mathcal{D}_i^j} |h(f(x,a)) - a|$$

Equality of Odds: $P(\hat{Y} = 1|A = 0, Y) = P(\hat{Y} = 1|A = 1, Y)$



[Madras et al, 2018](#)

Discrimination Measures for Representations

$$\mathcal{Z}_1 = p(Z|A = 1) \quad \mathcal{Z}_0 = p(Z|A = 0) \quad \mathcal{Z}_a^y = p(Z|A = a, Y = y)$$

- Demographic Parity

$$\Delta_{DP}(g) \triangleq d_g(\mathcal{Z}_0, \mathcal{Z}_1) = |\mathbb{E}_{\mathcal{Z}_0}[g] - \mathbb{E}_{\mathcal{Z}_1}[g]|$$

$$\text{Demographic Parity: } P(\hat{Y} = 1|A = 1) = P(\hat{Y} = 1|A = 0)$$

- Equality of Odds

$$\Delta_{EO}(g) \triangleq |\mathbb{E}_{\mathcal{Z}_0^0}[g] - \mathbb{E}_{\mathcal{Z}_1^0}[g]| + |\mathbb{E}_{\mathcal{Z}_0^1}[1 - g] - \mathbb{E}_{\mathcal{Z}_1^1}[1 - g]|$$

$$\text{Equality of Odds: } P(\hat{Y} = 1|A = 0, Y) = P(\hat{Y} = 1|A = 1, Y)$$

- Equality of Opportunities

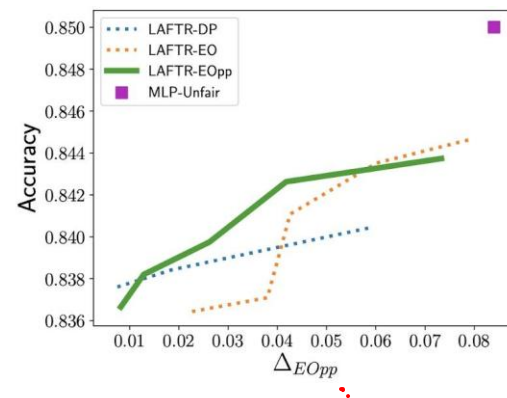
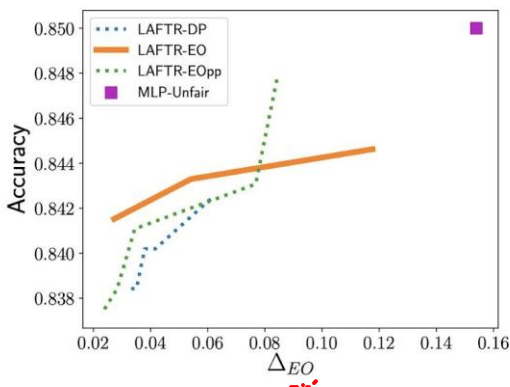
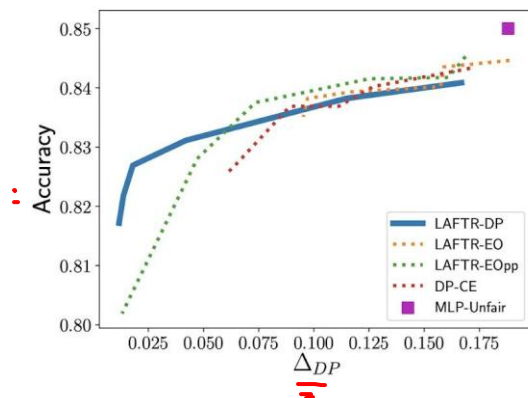
$$\Delta_{EOpp}(g) \triangleq |\mathbb{E}_{\mathcal{Z}_0^0}[g] - \mathbb{E}_{\mathcal{Z}_1^0}[g]|$$

$$\text{Equality of Opportunity: } P(\hat{Y} = 1|A = 0, Y = 1) = P(\hat{Y} = 1|A = 1, Y = 1)$$

Accuracy and Fairness on Adult Income Dataset

- Results Generated By Varying γ

$$L(f, g, h, k) = \alpha L_C(g(f(X, A)), Y) + \beta L_{Dec}(k(f(X, A), A), X) + \gamma L_{Adv}(h(f(X, A)), A)$$



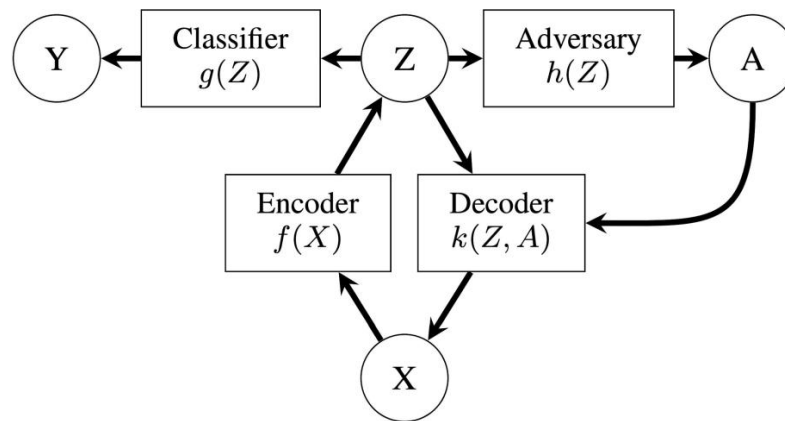
[Madras et al, 2018](#)

DP-CE - Cross Entropy Adversarial Objective ([Edwards et al, 2016](#))

Transferring Fair Representations



- If the Representations Are Fair, All Predictors Should Be Fair!
 - Train f and g based on domain 1 with feature space X
 - Fix f , and train g' on domain 2 with the same feature space X
 - $y=g'(f(x))$ should be a fairness predictor



[Madras et al, 2018](#)

In a nutshell

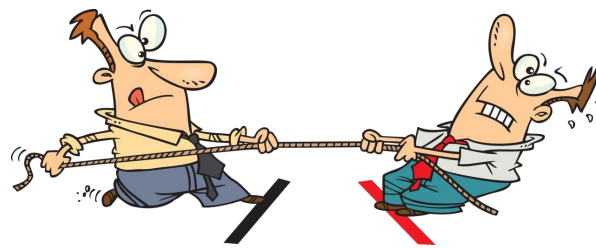
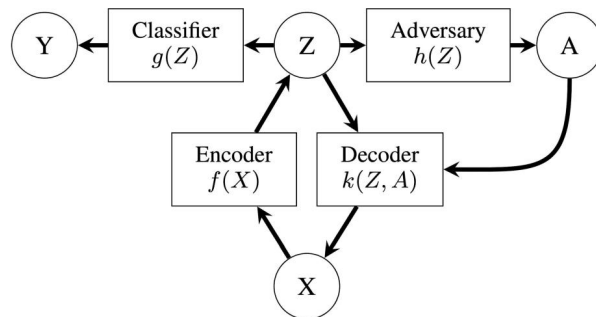


- Fair Representation Learning
 - Prejudice Removing Regularizer

$$-\mathcal{L}(\mathcal{D}; \boldsymbol{\Theta}) + \eta R(\mathcal{D}, \boldsymbol{\Theta}) + \frac{\lambda}{2} \|\boldsymbol{\Theta}\|_2^2$$

Loss of the Model Fairness Regularizer L2 Regularizer

- Fair Representations Through Adversarial Learning



Model Prediction ↑

Reconstruct A ↓

Comparisons: Regularization and Adversarial Learning

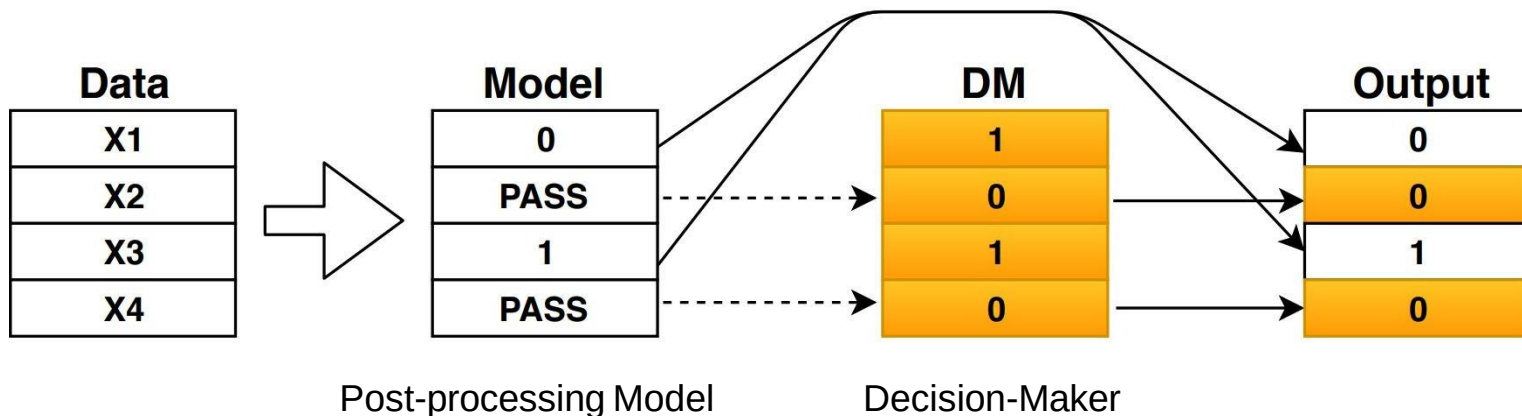


	Prejudice Removing Regularizer	Adversarial Learning
Pros	Minimal modifications to training procedure	Transferable representations
		Can be applied to many different fairness criteria
Cons	Can only be applied to Demographic Parity	Adversarial loss can be difficult to train

Post-Processing Methods for Fairness



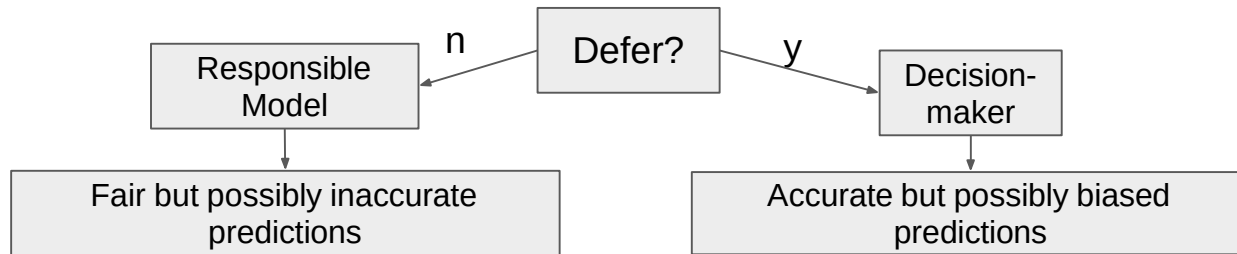
- Why Post-Processing?
 - Flexibility: No need to fine-tune the ML model
 - Model Agnostic: Can be applied across a wide range of models
- Learning to Defer



Learning to Defer



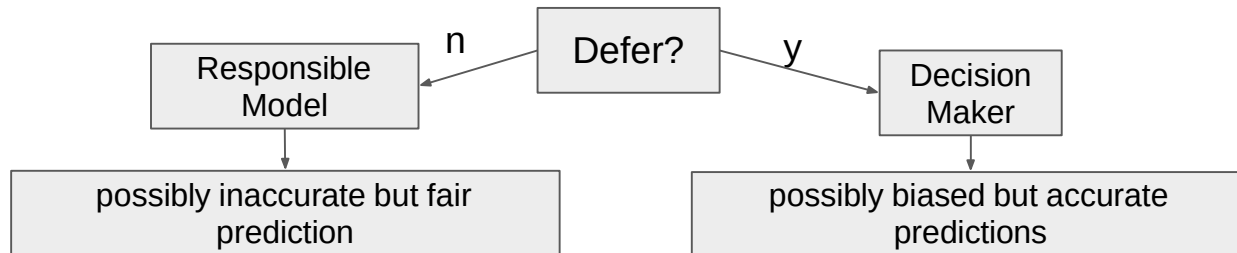
- Working Together with A Black-box Decision-maker Model
 - Decision-maker models (e.g. human) have access to important information that our model does not has
 - Decision-maker models might be biased
- Performance and Fairness Trade-offs
 - Fix the unfair predictions of the decision-maker model
 - Defer to the decision-maker the model is uncertain



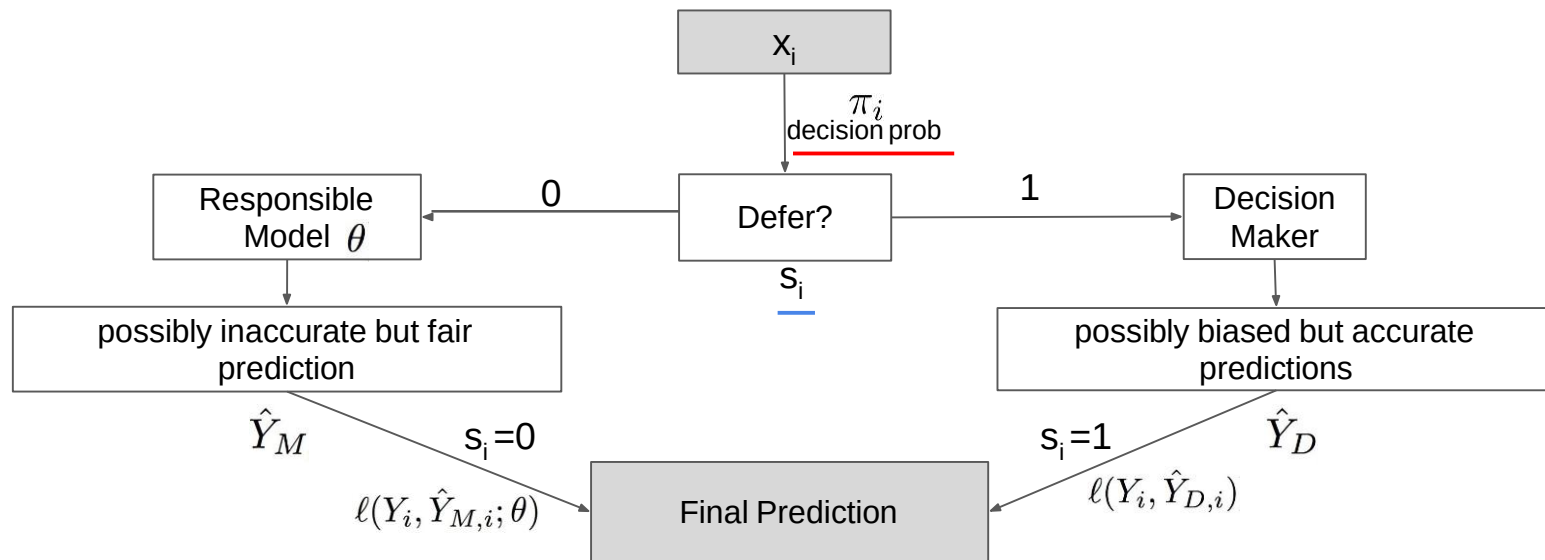
Learning to Defer



- Decision-maker Model
 - Considered as a black-box model
 - No fine-tuning, no access to its training data
- Responsible Model
 - Have access to additional data
 - Stick to fairness constraints



Training the Defer Model



$$\mathcal{L}_{defer}(Y, \hat{Y}_M, \hat{Y}_D, \pi; \theta) = \sum_i \mathbb{E}_{s_i \sim \text{Ber}(\pi_i)} [(1 - s_i) \ell(Y_i, \hat{Y}_{M,i}; \theta) + s_i \ell(Y_i, \hat{Y}_{D,i})] + \alpha_{fair} \mathcal{R}(Y, \hat{Y}_M, \hat{Y}_D, s)$$

Fair regularizer

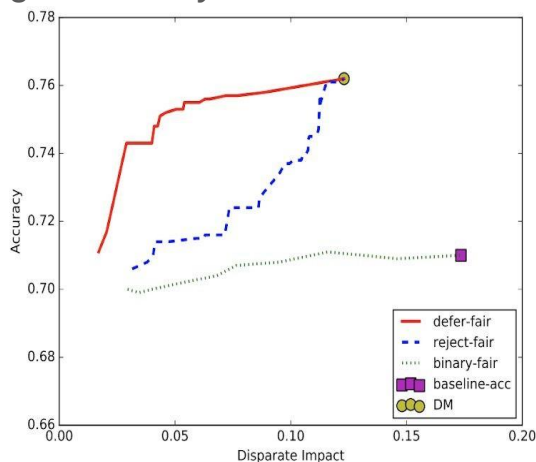
[Madras et al, 2018](#)

Results on COMPAS



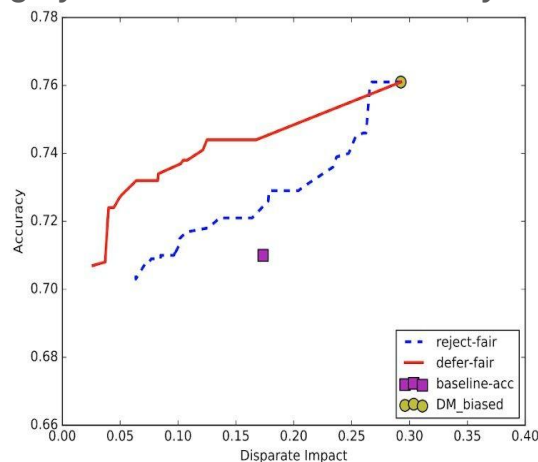
- DM Model

- High-Accuracy - DM has more data, Highly-Biased - DM is extremely biased



COMPAS, High-Accuracy DM

- DM - Decision-maker model
- Defer - Fair - Learning to Defer
- Reject- Fair - Only reject or accept DM



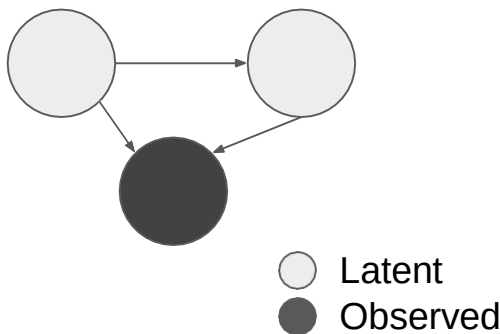
COMPAS, Highly-Biased DM

- Baseline - Model trained only to optimize accuracy, no DM
- Binary - Fair - Baseline optimized with fairness

Next Lecture: Fair Causal Reasoning

Causal Graph

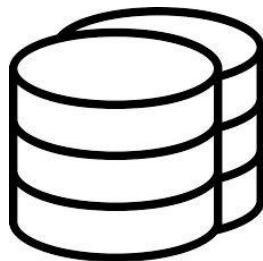
- Observed Data
- Latent Data
- Relations



Causal Fairness Criteria

- Counterfactual Fairness

Observed Data



Observational Fairness Criteria

- Fairness Through Unawareness
- Demographic Parity
- Equalized Odds/Opp